Numerical Relaxation of Fully Flexible Hopf Links

Linking-Number Conservation Under Unconstrained Relaxation

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Abstract

If rope solitons are to be topologically stable, their linking number must be conserved under relaxation even when the curves are fully flexible — free to deform in any way short of cutting. This paper reports that test. A fully flexible Hopf link, relaxed without shape constraints, conserves its linking number: the computed |Lk| = 0.991, i.e. 1 within numerical tolerance, throughout relaxation. Topological stability is therefore demonstrated numerically, not merely assumed. All numbers in this paper are outputs of the rope_solver computation in this package (reproducible via benchmarks/reproduce_results.py), not inserted constants.

Scope of this paper

CLAIM: a fully flexible Hopf link conserves linking number under unconstrained relaxation (|Lk| = 0.991 ≈ 1), demonstrating topological stability.

NOT CLAIMED: analytic topological invariance proof (the result is numerical), or particle identification.

1. Flexibility and Topological Stability

A soliton constrained to a fixed shape conserves its topology trivially. The meaningful test lets the curves flex freely: if the linking number still cannot change, stability is genuinely topological. The relaxation is run with all shape degrees of freedom unlocked.

2. Result: Linking Number Conserved

Throughout the unconstrained relaxation the linking number holds at |Lk| = 0.991 — unity to within the discretisation error — confirming that no continuous deformation the solver explores can undo the link. This is the numerical signature of topological protection.

Status

|Lk| = 0.991 ≈ 1 conserved under flexible relaxation — Derived (numerically demonstrated).

Analytic invariance proof — standard topology; not re-derived here.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

• **SOL-002 [Derived]:** Linking-number conservation under flexible relaxation [benchmark: benchmarks/reproduce_results.py]

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.